

Worksheet 4

Problem 1. Linear Functionals. Let $v_1, \dots, v_n$ be a basis for a real vector space V , and let $\varphi_1, \dots, \varphi_n$ be associated dual basis. Fix some integer $0 < m \leq n$, and define a function $T : V \rightarrow \mathbb{R}^m$ by the rule

$$T(v) = \begin{bmatrix} \varphi_1(v) \\ \varphi_2(v) \\ \vdots \\ \varphi_m(v) \end{bmatrix}.$$

- (a) Prove that T is a linear transformation.
- (b) Determine the rank of T .
- (c) Use part (b) to quickly find a basis for the nullspace of T .

Solution (a). We must check that T satisfies the two axioms (additivity and homogeneity) of a linear transformation.

$$T(v + w) = \begin{bmatrix} \varphi_1(v + w) \\ \varphi_2(v + w) \\ \vdots \\ \varphi_m(v + w) \end{bmatrix} = \begin{bmatrix} \varphi_1(v) \\ \varphi_2(v) \\ \vdots \\ \varphi_m(v) \end{bmatrix} + \begin{bmatrix} \varphi_1(w) \\ \varphi_2(w) \\ \vdots \\ \varphi_m(w) \end{bmatrix} = T(v) + T(w).$$

$$T(\lambda v) = \begin{bmatrix} \varphi_1(\lambda v) \\ \varphi_2(\lambda v) \\ \vdots \\ \varphi_m(\lambda v) \end{bmatrix} = \begin{bmatrix} \lambda \varphi_1(v) \\ \lambda \varphi_2(v) \\ \vdots \\ \lambda \varphi_m(v) \end{bmatrix} = \lambda T(v).$$

Solution (b). Recall that the dual basis has the following property: $\varphi_i(v_j) = 0$ if $i \neq j$ and $\varphi_i(v_j) = 1$ if $i = j$. Substituting these values into our expression for T , we find that $T(v_i) = e_i$ for $1 \leq i \leq m$ and $T(v_i) = 0$ for $i > m$. As a consequence, the standard basis for $\mathbb{R}^m$ belongs to the range of T , so T has the maximum possible rank m .

Solution (c). By rank-nullity, the kernel of T has dimension $n - m$. To demonstrate a basis, it suffices to find $n - m$ linearly independent vectors in the nullspace of T . By reasoning in part (b), we find that the vectors $v_{m+1}, \dots, v_n$ all live in the nullspace of T . Since they are part of a basis for V , these vectors are independent and therefore form a basis for $\text{null}(T)$.

Problem 2. Quotient Spaces. Let $V = \mathcal{P}_3(\mathbb{R})$ be the space of real cubic polynomials. Define a linear transformation $T : V \rightarrow V$ by the rule

$$T(p) = p''.$$

That is, T is the linear transformation sending a polynomial to its second derivative.

- (a) Check that the subspace

$$U = \text{span}(1, x)$$

is the nullspace of T .

- (b) Describe the quotient space V/U . What do vectors in this space look like? What does addition look like? What is its dimension?
- (c) Prove that $\hat{T} : V/U \rightarrow V/U$ is the zero transformation, that is, it sends everything to the zero vector.

Solution (a). The second derivative of any linear polynomial is zero. The second derivative of any degree two or more polynomial is nonzero. Thus, the kernel of T is exactly the set of linear polynomials. Another way of describing this set is as a span of 1 and x .

Solution (b). Recall that the quotient space V/U consists of elements $[p]$ where $p \in V$ is some representative. We say two elements $[p], [q]$ are equal if $p + U = q + U$. Since U consists of linear polynomials, we notice that $a_0 + a_1x + U = 0 + U$ for any linear polynomial $a_0 + a_1x$. In particular, this implies

$$[a_0 + a_1x + a_2x^2 + a_3x^3] = [a_0 + a_1x] + [a_2x^2 + a_3x^3] = [a_2x^2 + a_3x^3] = a_2[x^2] + a_3[x^3].$$

We now claim that the vectors $[x^2]$ and $[x^3]$ are linearly independent, in which case they must form a basis for V/U . The claim follows immediately after observing that

$$ax^2 + U \neq bx^3 + U$$

for any scalars a, b . The quotient V/U is two-dimensional, and elements are identified as degree three polynomials missing the constant and linear terms.

Solution (c). Because U is in the nullspace of T , there is a well defined linear transformation $\bar{T} : V/U \rightarrow V$ given by $\bar{T}([p]) = T(p)$. We can further quotient by U to define a transformation $\hat{T} : V/U \rightarrow V/U$ by the rule $\hat{T}([p]) = [T(p)]$. The claim that T is the zero transformation is equivalent to saying that $T(p) \in U$ for all polynomials p . Indeed, this is true since the second derivative of a degree three or less polynomial has degree one or less and lives in U .

Problem 3. Linear Operators. Let $T : V \rightarrow V$ be a linear transformation from some vector space V back to itself. We define the *square* $T^2 : V \rightarrow V$ as the function $T^2(v) = T(T(v))$.

- (a) Prove that T^2 is a linear function.
- (b) Show that $\text{null}(T) \subset \text{null}(T^2)$.
- (c) Show that $\text{range}(T^2) \subset \text{range}(T)$.

Solution (a). We must check the additivity and homogeneity of the square transformation;

$$T^2(v + w) = T(T(v + w)) = T(T(v) + T(w)) = T(T(v)) + T(T(w)) = T^2(v) + T^2(w).$$

$$T^2(\lambda v) = T(T(\lambda v)) = T(\lambda T(v)) = \lambda T(T(v)) = \lambda T^2(v).$$

Solution (b). Let v be a vector in the nullspace of T . This means that $T(v) = 0$. Applying the square transformation gives $T^2(v) = T(T(v)) = T(0) = 0$ since every linear transformation sends the zero vector to the zero vector. Thus, $v \in \text{null}(T)$, and the inclusion in part (b) holds true.

Solution (c). Recall that the range of T^2 is the set of elements having the form $T^2(v)$ for some $v \in V$. Rewriting the square function with composition yields $T^2(v) = T(T(v))$. The element $T(T(v))$ belongs to the range of T , thereby giving the inclusion in (c).